

Category of courses:**A. Language and General Education****1. Language**

Sl. No.	Course Code	Course Title	Credits
1	0231-101	Communicative English	3.0
2	0232-402	Technical Writing and Presentation Lab	1.0
Total Credits			4.0

2. Arts and Humanities

Sl. No.	Course Code	Course Title	Credits
1	0222-101	Technology and Society	3.0
Total Credits			3.0

3. Social Science

Sl. No.	Course Code	Course Title	Credits
1	0223-201	Professional Ethics and Environmental Protection	3.0
2	0311-101	Economics	3.0
Total Credits			6.0

4. Business

Sl. No.	Course Code	Course Title	Credits
1	0413-201	Information System Management	3.0
2	0413-401	Project Management and Finance	3.0
3	0413-402	Software Project Management Lab	1.0
4	0413-403	Technopreneurship	3.0
Total Credits			10.0

B. Basic Sciences

Sl. No.	Course Code	Course Title	Credits
1	0533-101	Physics I	3.0
2	0533-102	Physics I Lab	1.0
3	0533-103	Physics II	3.0
Total Credits			7.0

C. Mathematics

Sl. No.	Course Code	Course Title	Credits
1	0541-101	Differential and Integral Calculus and Vector Analysis	3.0
2	0541-102	Linear Algebra and Complex Variables	3.0
3	0541-201	Discrete Mathematics	3.0
4	0541-202	Numerical Analysis Lab	1.0
5	0542-201	Statistical Methods and Probability	3.0
6	0542-202	Statistical Computing Lab	1.0
Total Credits			14.0

D. Other Engineering

Sl. No.	Course Code	Course Title	Credits
1	0713-101	Electronic Devices and Circuits	3.0
2	0713-102	Electronic Devices and Circuits Lab	1.0
3	0713-201	Digital Logic Design	3.0
4	0713-202	Digital Logic Design Lab	1.0
Total Credits			8.0

E. Computer Science and Engineering Core Courses

1. Programming

Sl. No.	Course Code	Course Title	Credits
1	0613-101	Structured Programming Language	3.0
2	0613-102	Structured Programming Language Lab	1.0
3	0613-201	Object Oriented Programming	3.0
4	0613-202	Object Oriented Programming Lab	1.0
5	0613-301	Web Programming	3.0
6	0613-302	Web Programming Lab	1.0
7	0613-310	Mobile Application Development Lab	1.0
Total Credits			13.0

2. Logic and Algorithms

Sl. No.	Course Code	Course Title	Credits
1	0613-103	Data Structures and Algorithms I	3.0
2	0613-104	Data Structures and Algorithms I Lab	1.0
3	0613-203	Data Structures and Algorithms II	3.0
4	0613-204	Data Structures and Algorithms II Lab	1.0
5	0613-309	Theory of Computation	3.0
6	0613-313	Artificial Intelligence	3.0

7	0613-314	Artificial Intelligence Lab	1.0
8	0613-401	Machine Learning	3.0
9	0613-402	Machine Learning Lab	1.0
Total Credits			19.0

3. Software Systems and Engineering

Sl. No.	Course Code	Course Title	Credits
1	0613-303	Software Engineering	3.0
2	0613-304	Software Development with .NET Framework	1.0
3	0613-306	Software Testing and Quality Assurance Lab	1.0
Total Credits			5.0

4. Hardware Systems

Sl. No.	Course Code	Course Title	Credits
1	0613-305	Computer Organization and Architecture	3.0
2	0613-307	Microprocessor, Microcontroller and Assembly Language	3.0
3	0613-308	Microprocessor, Microcontroller and Assembly Language Lab	1.0
Total Credits			7.0

5. Systems

Sl. No.	Course Code	Course Title	Credits
1	0612-201	Database Management System	3.0
2	0612-202	Database Management System Lab	1.0
3	0612-203	Data Communication	3.0
4	0612-204	Data Communication Lab	1.0
5	0612-301	Computer Networking	3.0
6	0612-302	Computer Networking Lab	1.0
7	0612-401	Information Security Fundamentals	3.0
8	0612-402	Information Security Fundamentals Lab	1.0
9	0613-311	Operating System	3.0
10	0613-312	Operating System Lab	1.0
Total Credits			20.0

6. Project Work

Sl. No.	Course Code	Course Title	Credits
1	0688-400	Capstone Project Design	6.0
Total Credits			6.0

F. Elective Courses

1. Theoretical Computer Science

Sl. No.	Course Code	Course Title	Credits
1	0541-401	Mathematical Analysis for Computer Science	3.0
2	0613-403	Graph Theory	3.0
3	0613-404	Computational Geometry	3.0
4	0613-405	Parallel Algorithms	3.0
5	0613-406	Simulation and Modeling	3.0
6	0613-407	Compiler Design	3.0
7	0613-408	Computer Graphics and Multimedia	3.0

2. Communication and Networking

Sl. No.	Course Code	Course Title	Credits
1	0612-403	Cloud Computing	3.0
2	0612-404	Satellite Communication Engineering	3.0
3	0612-405	Wireless Networks	3.0
4	0612-406	System and Network Administration	3.0

3. Software Engineering

Sl. No.	Course Code	Course Title	Credits
1	0612-407	High-Performance Database Management System	3.0
2	0613-409	Advanced Programming with Python	3.0
3	0613-410	Enterprise Resource Planning	3.0

4. Artificial Intelligence and Data Science

Sl. No.	Course Code	Course Title	Credits
2	0613-411	Pattern Recognition	3.0
3	0613-412	Digital Image and Video Processing	3.0
4	0613-413	Natural Language Processing	3.0
5	0613-414	Bioinformatics	3.0
6	0612-408	Data and Web Mining	3.0
7	0612-409	Big Data Analytics	3.0
8	0612-410	Data Science and Application	3.0
9	0612-411	Complex Data Visualization	3.0

5. Hardware System

Sl. No.	Course Code	Course Title	Credits
1	0611-401	Human Machine Interaction	3.0
2	0714-401	Robotics	3.0
3	0714-402	Internet of Things	3.0
4	0714-403	Microprocessor Interfacing	3.0
5	0714-404	Very Large Scale Integration	3.0
6	0613-415	Quantum Computing	3.0

6. Cyber Security

Sl. No.	Course Code	Course Title	Credits
1	0612-412	Blockchain and Distributed Security	3.0
2	0612-413	Web Application Security	3.0
3	0612-414	Ethical Hacking and System Security	3.0
4	0612-415	Digital Forensics and Incident Response	3.0
5	0612-416	Cryptography and Cryptoanalysis	3.0
6	0612-417	Network Security	3.0
7	0612-418	Computer and Cybersecurity	3.0

7. ICT

Sl. No.	Course Code	Course Title	Credits
1	0413-404	Enterprise Systems: Concepts and Practice	3.0
2	0413-405	Electronic Business	3.0
3	0413-406	Business Analytics	3.0
4	0413-407	Entrepreneurship: Innovation and Commercialization	3.0
5	0611-402	ICT for Development	3.0
6	0611-403	IT Audit: Concepts and Practice	3.0

18.1 Year and Semester-wise Distribution of Courses

(a) First Year First Semester

Sl. No.	Course Code	Course Title	Credit
1	0222-101	Technology and Society	3.0
2	0231-101	Communicative English	3.0
3	0533-101	Physics I	3.0
4	0533-102	Physics I Lab	1.0
5	0541-101	Differential and Integral Calculus and Vector Analysis	3.0
6	0613-101	Structured Programming Language	3.0
7	0613-102	Structured Programming Language Lab	1.0
			Total Credits = 17

(b) First Year Second Semester

Sl. No.	Course Code	Course Title	Credit
1	0311-101	Economics	3.0
2	0533-103	Physics II	3.0
3	0541-102	Linear Algebra and Complex Variables	3.0
4	0613-103	Data Structures and Algorithms I	3.0
5	0613-104	Data Structures and Algorithms I Lab	1.0
6	0713-101	Electronic Devices and Circuits	3.0
7	0713-102	Electronic Devices and Circuits Lab	1.0
			Credits = 17

(c) Second Year First Semester

Sl. No.	Course Code	Course Title	Credit
1	0223-201	Professional Ethics and Environmental Protection	3.0
2	0541-201	Discrete Mathematics	3.0
3	0542-201	Statistical Methods and Probability	3.0
4	0542-202	Statistical Computing Lab	1.0
5	0613-201	Object Oriented Programming	3.0
6	0613-202	Object Oriented Programming Lab	1.0
7	0713-201	Digital Logic Design	3.0
8	0713-202	Digital Logic Design Lab	1.0
			Total Credits = 18

(d) Second Year Second Semester

Sl. No.	Course Code	Course Title	Credit
1	0413-201	Information System Management	3.0
2	0541-202	Numerical Analysis Lab	1.0
3	0612-201	Database Management System	3.0
4	0612-202	Database Management System Lab	1.0
5	0612-203	Data Communication	3.0
6	0612-204	Data Communication Lab	1.0
7	0613-203	Data Structures and Algorithms II	3.0
8	0613-204	Data Structures and Algorithms II Lab	1.0
			Total Credits=16

(e) Third Year First Semester

Sl. No.	Course Code	Course Title	Credit
1	0612-301	Computer Networking	3.0
2	0612-302	Computer Networking Lab	1.0
3	0613-301	Web Programming	3.0
4	0613-302	Web Programming Lab	1.0
5	0613-303	Software Engineering	3.0
6	0613-304	Software Development with .Net Framework	1.0
7	0613-305	Computer Organization and Architecture	3.0
8	0613-306	Software Testing and Quality Assurance Lab	1.0
			Total Credits = 16

(f) Third Year Second Semester

Sl. No.	Course Code	Course Title	Credit
1	0613-307	Microprocessor, Microcontroller and Assembly Language	3.0
2	0613-308	Microprocessor, Microcontroller and Assembly Language Lab	1.0
3	0613-309	Theory of Computation	3.0
4	0613-310	Mobile Application Development Lab	1.0
5	0613-311	Operating System	3.0
6	0613-312	Operating System Lab	1.0

7	0613-313	Artificial Intelligence	3.0
8	0613-314	Artificial Intelligence Lab	1.0
	Total Credits = 16		

(g) Fourth Year First Semester

Sl. No.	Course Code	Course Title	Credit
1	0232-402	Technical Writing and Presentation Lab	1.0
2	0413-401	Project Management and Finance	3.0
3	0413-402	Software Project Management Lab	1.0
4	0613-401	Machine Learning	3.0
5	0613-402	Machine Learning Lab	1.0
6	0541/0413/0612/0613/0619/0714-4**	Elective Course I	3.0
7	0541/0413/0612/0613/0619/0714-4**	Elective Course II	3.0
8	0541/0413/0612/0613/0619/0714-4**	Elective Course III	3.0
9	0688-400	Capstone Project Design I	2.0
			Total Credits = 20

(a) Fourth Year Second Semester

Sl. No.	Course Code	Course Title	Credit
1	0413-403	Technopreneurship	3.0
2	0612-401	Information Security Fundamentals	3.0
3	0612-402	Information Security Fundamentals Lab	1.0
4	0541/0413/0612/0613/0619/0714-4**	Elective Course IV	3.0
5	0541/0413/0612/0613/0619/0714-4**	Elective Course V	3.0
6	0541/0413/0612/0613/0619/0714-4**	Elective Course VI	3.0
7	0688-400	Capstone Project Design II	4.0
			Total Credits = 20